Project Synopsis

1. Title

Working Project Title

Wikipedia Attention Shock Forecaster Using Time-Series Analysis

Team Details

Student Name	College Name	Branch	Mobile No	Mail ID	Role (Lead/Member)
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Program Track and Domain

Program Track (Engg-HUB / Engg-Spoke / Science-HUB / Science-Spoke / Commerce-Spoke)	Domain (Analytics / Machine Learning / Deep Learning / GenAI)
Engg-HUB	Machine Learning / Analytics

Guide Details

Aditya Ardak (Trainer)

2. Problem Statement

Wikipedia pageviews are a strong indicator of public interest in real-world topics, events, and personalities. Whenever a major event occurs such as breaking news, viral trends, or important announcements people often visit Wikipedia to learn more, leading to sudden spikes in pageviews. These spikes reflect rapid changes in collective attention, but there is no simple system that can automatically identify and analyze such sudden increases in a structured way.

This project aims to address this problem by developing a system that forecasts expected pageviews using time-series techniques and detects abnormal spikes, known as "attention shocks." By comparing predicted and actual values, the system highlights unusual patterns and visualizes them through an interactive Streamlit dashboard. This helps researchers, analysts, and content creators understand how public interest changes over time and respond to real-world events more effectively.

3. Objectives

The main objectives of this project are:

1. To collect Wikipedia pageview data using an API for selected articles and time ranges.

2. To preprocess and organize the data into a proper time-series format by handling missing values and inconsistencies.

3. To develop a forecasting model that can predict expected pageviews based on historical data patterns.

4. To identify and detect abnormal spikes in pageviews, referred to as "attention shocks," using analytical techniques.

5. To visualize pageview trends and detected shocks through clear and informative graphs.

6. To design and develop an interactive dashboard using Streamlit for easy user interaction and analysis.

4. Methodology

The proposed system follows a structured workflow to forecast Wikipedia pageviews and detect attention shocks using time-series analysis.

Step 1: Data Collection

Wikipedia pageview data will be collected using the Wikimedia Pageviews API. The data includes daily pageviews of selected articles over a specified time range.

Step 2: Data Preprocessing

The collected data will be cleaned by handling missing values, removing inconsistencies, and converting the data into a proper time-series format. Date fields will be formatted, and gaps in data will be managed appropriately.

Step 3: Feature Engineering

Relevant features such as rolling mean, rolling standard deviation, and trend-based indicators will be created to capture patterns and variations in pageviews over time.

Step 4: Model Training

A time-series forecasting model such as ARIMA/SARIMAX or Exponential Smoothing will be used to learn patterns from historical data and predict expected pageviews.

Step 5: Shock Detection

The difference between actual and predicted values (residuals) will be calculated. Abnormal spikes, known as attention shocks, will be detected using threshold-based or statistical methods.

Step 6: Model Evaluation

The performance of the model will be evaluated using metrics such as:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Mean Absolute Percentage Error (MAPE)

Step 7: Deployment

An interactive web-based dashboard will be developed using Streamlit to visualize trends, forecasts, and detected attention shocks.

5. Resources and Timeline

A. Hardware & Software Requirements

Hardware

•Laptop / PC

• Minimum 8 GB RAM

• Processor: Intel i5 or equivalent

Software

• Operating System: Windows / Linux

• Programming Language: Python

• Development Tools:

- Jupyter Notebook
- VS Code
- Streamlit

• Libraries:

- Pandas
- NumPy
- Statsmodels
- Matplotlib
- Plotly

B. Dataset

Source: Wikimedia Pageviews API (https://doc.wikimedia.org/generated-data-platform/aqs/analytics-api/reference/page-views.html)

Type: Time-series data

Data: Daily pageviews of Wikipedia articles

The dataset includes the Wikipedia article name, the date of each entry, and the number of daily pageviews.

C. Project Timeline

Phase	Task	Tentative Completion
Phase 1	Data Collection from Wikimedia Pageviews API	Week 1
Phase 2	Data Preprocessing (cleaning, formatting, handling missing values)	Week 1
Phase 3	Feature Engineering (rolling mean, std, trend indicators)	Week 1
Phase 4	Model Training (ARIMA/SARIMAX or Exponential Smoothing)	Week 1
Phase 5	Shock Detection (identify attention spikes from residuals)	Week 1
Phase 6	Model Evaluation (MAE, RMSE, MAPE)	Week 1
Phase 7	Deployment (Interactive dashboard with Streamlit)	Week 1

6. Expected Outcomes / Conclusion

The proposed system will provide an end-to-end solution for forecasting Wikipedia pageviews and detecting sudden attention spikes. The expected outcomes include:

- **Working Forecasting Model:** A reliable time-series model (ARIMA/SARIMAX or Exponential Smoothing) to predict future pageviews accurately.
- **Attention Spike Detection:** Identification of abnormal surges in pageviews, helping to track viral trends or sudden public interest.
- **Interactive Visualizations:** Clear charts and graphs to show trends, seasonal patterns, and detected shocks.
- **Functional Streamlit Dashboard:** A user-friendly web interface to display forecasts, anomalies, and visualizations interactively.
- **Real-Time Analysis Capability:** Continuous monitoring using live data from Wikimedia Pageviews API for timely insights.

7. References

1. 1. Wikimedia Analytics API Documentation

https://doc.wikimedia.org/generated-data-platform/aqs/analytics-api/

2. Pageviews API Reference

https://doc.wikimedia.org/generated-data-platform/aqs/analytics-api/reference/page-views.html

3. Pageviews Analysis Tool

https://pageviews.wmcloud.org/

4. MediaWiki REST API Documentation

https://www.mediawiki.org/wiki/API:REST_API

5. Pandas Documentation

https://pandas.pydata.org/docs/

6. Scikit-learn Documentation

https://scikit-learn.org/

7. Streamlit Documentation

https://docs.streamlit.io/

8. Research on Wikipedia Usage Patterns and Trends